Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Linear Graphs 5: Parallel Lines

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

KEY WORDS gradient *(rōnaki)* · y-intercept *(pūtahi-y)* · y = mx + c · two points · parallel *(whakarara)*

A Gradient from two points ●○○

Remember: Gradient $m = \dfrac{y_2 - y_1}{x_2 - x_1}$. Keep the points in the same order on the top and the bottom.

1. Find the gradient of the line through each pair of points.

a) $(1,\ 3)$ and $(4,\ 9)$

b) $(0,\ 5)$ and $(2,\ 1)$

c) $(-2,\ 1)$ and $(4,\ 4)$

d) $(3,\ -2)$ and $(5,\ 8)$

e) $(-1,\ 6)$ and $(3,\ -2)$

f) $(-4,\ -3)$ and $(2,\ 0)$

B From a gradient and a point ●●○

Remember: Put the gradient and the point into $y = mx + c$ and solve for c. Then write the equation.

2. Find the equation of the line with the given gradient that passes through the given point.

a) Gradient 2, through $(1,\ 5)$

b) Gradient 3, through $(2,\ 1)$

c) Gradient -1, through $(4,\ 2)$

d) Gradient -2, through $(-1,\ 6)$

e) Gradient $\dfrac{1}{2}$, through $(4,\ 5)$

f) Gradient $-\dfrac{2}{3}$, through $(3,\ -1)$

C From two points ●●○

Remember: Find the gradient first. Then use either point to find c. Check your equation with the other point.

3. Find the equation of the line through each pair of points.

a) $(1,\ 3)$ and $(3,\ 7)$

b) $(0,\ 5)$ and $(2,\ 1)$

c) $(2,\ 1)$ and $(4,\ 2)$

d) $(-1,\ -5)$ and $(2,\ 4)$

e) $(-2,\ 6)$ and $(2,\ -2)$

f) $(-3,\ -1)$ and $(3,\ 3)$

D Parallel lines ●●●

Remember: Parallel lines have the same gradient. If an equation is not in the form $y = mx + c$, rearrange it before you read off the gradient.

4. Find the equation of each line.

a) Parallel to $y = 3x - 1$, through $(2,\ 10)$

b) Parallel to $y = -2x + 5$, through $(-1,\ 4)$

c) Parallel to $2y = 6x + 2$, through $(1,\ 2)$

d) Parallel to $x + 2y = 6$, through $(4,\ -1)$

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

5. Are these pairs of lines parallel? Show how you know.

a) $y = 2x + 3$ and $2y = 4x - 2$

b) $3x + y = 7$ and $y = 3x - 2$

c) $x - 2y = 4$ and $2y = x + 6$

d) $y = 4 - x$ and $x + y = 1$

E Checking points and finding intercepts ●●●

> **Remember:** To test a point, put its x into the equation and see whether you get its y. A line crosses the x-axis where $y = 0$.

6. Is each point on the line? Show your working.

a) Is $(4,\ 11)$ on the line $y = 2x + 3$?

b) Is $(2,\ 1)$ on the line $y = -3x + 5$?

c) Is $(6,\ -1)$ on the line $y = \frac{1}{2}x - 4$?

7. Find the equation of the line through each pair of points, then the point where it crosses the x-axis.

a) $(1,\ 1)$ and $(3,\ 5)$

b) $(-2,\ 8)$ and $(2,\ 0)$

c) $(0,\ 3)$ and $(6,\ 6)$

Extension

8. Two lines are perpendicular (at right angles) when their gradients multiply to -1. A line has gradient 2. What is the gradient of a line perpendicular to it?

9. Find the equation of the line through $(2,\ 1)$ that is perpendicular to $y = \frac{1}{2}x + 3$.

10. A is $(1,\ 2)$ and B is $(5,\ 4)$.

a) Find the midpoint of AB.

b) Find the equation of the perpendicular bisector of AB: the line through the midpoint at right angles to AB.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Linear Graphs 5: Parallel Lines

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS gradient *(rōnaki)* · y-intercept *(pūtahi-y)* · y = mx + c · two points · parallel *(whakarara)*

A Gradient from two points ●○○

Remember: Gradient $m = \dfrac{y_2 - y_1}{x_2 - x_1}$. Keep the points in the same order on the top and the bottom.

1. Find the gradient of the line through each pair of points.

a) $(1,\ 2)$ and $(3,\ 10)$

b) $(0,\ 7)$ and $(3,\ 1)$

c) $(-3,\ 2)$ and $(3,\ 5)$

d) $(2,\ -5)$ and $(4,\ 7)$

e) $(-2,\ 5)$ and $(2,\ -3)$

f) $(-6,\ -1)$ and $(2,\ 3)$

B From a gradient and a point ●●○

Remember: Put the gradient and the point into $y = mx + c$ and solve for c. Then write the equation.

2. Find the equation of the line with the given gradient that passes through the given point.

a) Gradient 3, through $(1,\ 7)$

b) Gradient 2, through $(3,\ 1)$

c) Gradient -1, through $(5,\ 1)$

d) Gradient -3, through $(-1,\ 4)$

e) Gradient $\dfrac{1}{2}$, through $(6,\ 7)$

f) Gradient $-\dfrac{2}{3}$, through $(6,\ -2)$

C From two points ●●○

Remember: Find the gradient first. Then use either point to find c. Check your equation with the other point.

3. Find the equation of the line through each pair of points.

a) $(1,\ 5)$ and $(3,\ 11)$

b) $(0,\ 4)$ and $(3,\ -2)$

c) $(2,\ 3)$ and $(6,\ 5)$

d) $(-1,\ -7)$ and $(2,\ 5)$

e) $(-3,\ 5)$ and $(1,\ 1)$

f) $(-3,\ 1)$ and $(3,\ 5)$

D Parallel lines ●●●

Remember: Parallel lines have the same gradient. If an equation is not in the form $y = mx + c$, rearrange it before you read off the gradient.

4. Find the equation of each line.

a) Parallel to $y = 2x + 7$, through $(3,\ 1)$

b) Parallel to $y = -3x - 2$, through $(-1,\ 5)$

c) Parallel to $2y = 4x - 6$, through $(2,\ 1)$

d) Parallel to $x + 3y = 9$, through $(6,\ -2)$

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

5. Are these pairs of lines parallel? Show how you know.

a) $y = 3x - 2$ and $3y = 9x + 6$

b) $2x + y = 5$ and $y = 2x + 1$

c) $x - 3y = 6$ and $3y = x + 9$

d) $y = 5 - x$ and $x + y = 2$

E Checking points and finding intercepts ●●●

> **Remember:** To test a point, put its x into the equation and see whether you get its y. A line crosses the x-axis where $y = 0$.

6. Is each point on the line? Show your working.

a) Is $(4,\ 10)$ on the line $y = 3x - 2$?

b) Is $(3,\ -1)$ on the line $y = -2x + 7$?

c) Is $(8,\ 7)$ on the line $y = \frac{1}{2}x + 3$?

7. Find the equation of the line through each pair of points, then the point where it crosses the x-axis.

a) $(1,\ 3)$ and $(3,\ 7)$

b) $(-1,\ 7)$ and $(3,\ -1)$

c) $(0,\ 2)$ and $(4,\ 4)$

Extension

8. Two lines are perpendicular (at right angles) when their gradients multiply to -1. A line has gradient 3. What is the gradient of a line perpendicular to it?

9. Find the equation of the line through $(3,\ 1)$ that is perpendicular to $y = \frac{1}{3}x + 2$.

10. A is $(2,\ 1)$ and B is $(6,\ 3)$.

a) Find the midpoint of AB.

b) Find the equation of the perpendicular bisector of AB: the line through the midpoint at right angles to AB.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.